Test Plan

1. Purpose

This test plan aims to define the scope, objectives, approach, and focus areas of testing for the Oral Assessment System (OAS). The main goal is to verify the correctness, stability, and security of fundamental system functionalities before integration with external systems (e.g., Canvas, SSO).

2. Scope

The testing will cover the following core modules:

- User account management (student/examiner/admin registration and login)
- Scheduling system (booking/rescheduling/canceling slots)
- Marking and feedback input by examiners
- CSV data import/export (student information, marks, feedback)
- Notification and reminder system
- Basic access control and data privacy validation

3. Out of Scope

- Full Canvas integration
- SSO (Single Sign-On) authentication
- Mobile and tablet adaptive UI
- Large-scale performance testing
- Security penetration testing

4. Test Types

- Unit Testing (for service, utility, and controller layers)
- Integration Testing (for user login, scheduling, and CSV import/export)
- Functional Testing (end-to-end feature flow testing)
- Basic Security Testing (authentication, access permission, visibility control)
- Usability Smoke Testing (navigation, booking flow, simple UX check)

5. Test Approach

Use JUnit 5 and Vitest for unit and integration testing.
Perform manual functional tests using a browser (Chrome/Edge).
Automate basic API tests via Postman collections or REST-assured scripts.
Verify data integrity through direct MySQL database queries where necessary.
Validate CSV upload/download manually at early stages.

6. Test Deliverables

- Unit Test Reports (JUnit reports in CI/CD pipeline)
- Manual Test Cases (Excel sheet or markdown files)
- Bug/Issue Reports (tracked in GitHub Issues)
- Test Coverage Report (optional for later stages)

7. Test Environment

Local Spring Boot server with MySQL database
Staging server
GitHub repository and CI/CD pipeline
Browser: Chrome latest version
Testing Tools (e.g. Postman, MySQL Workbench)

8. Risks & Mitigation

TBD